

Generic type of Ahlswede–Zhang style identities

Tran Dan Thu

*Faculty of Information Technology
 University of Science, 227 Nguyen van Cu
 Dist. 5, Ho Chi Minh City, Vietnam
 tdt@hcmus.edu.vn*

Received 4 January 2017

Accepted 13 July 2017

Published 10 August 2017

We introduce a generic identity based on the Ahlswede–Zhang (AZ) function, which can be considered as a template to establish AZ-style identities. Many AZ-style identities can be derived from the generic identity.

Keywords: Poset; LYM inequality; Ahlswede–Zhang identity; generic identity.

Mathematics Subject Classification 2010: 05D05

1. Introduction

Let $[n] = \{1, 2, \dots, n\}$, $2^{[n]}$ denote the family of all subsets of $[n]$, $\mathcal{P}_k$ be the all subsets of $[n]$ of size k , and $\emptyset$ be the empty set. If $\emptyset \neq \mathcal{F} \subseteq 2^{[n]}$ and $A \not\subseteq B$ for all $A, B \in \mathcal{F}$, $A \neq B$, then $\mathcal{F}$ is called a *Sperner family* or an *antichain*.

Theorem 1 (Lubell [8], Meshalkin [9], Yamamoto [16]). *If $\mathcal{A}$ is an antichain then*

$$\sum_{A \in \mathcal{A}} \frac{1}{\binom{n}{|A|}} \leq 1. \quad (1)$$

The inequality (1) is called LYM inequality, which can be rewritten as $\sum_{k=0}^n \frac{|\mathcal{A} \cap \mathcal{P}_k|}{\binom{n}{k}} \leq 1$.

For every $\emptyset \neq \mathcal{F} \subseteq 2^{[n]}$ we define the downset

$$\mathcal{D}(\mathcal{F}) = \{D \subseteq [n] : D \subseteq F \text{ for some } F \in \mathcal{F}\},$$

and the upset

$$\mathcal{U}(\mathcal{F}) = \{U \subseteq [n] : U \supseteq F \text{ for some } F \in \mathcal{F}\}.$$

Then we put $Z_{\mathcal{F}}(X) = \bigcap_{X \supseteq F \in \mathcal{F}} F$ for each $X \in \mathcal{U}(\mathcal{F})$, and define $Z^*(\cdot)$, the dual of $Z(\cdot)$ for each $X \in \mathcal{D}(\mathcal{F})$, which is $Z_{\mathcal{F}}^*(X) = \bigcup_{X \subseteq F \in \mathcal{F}} F$.

The $Z(\cdot)$ function (called Ahlswede–Zhang (*AZ function*)) is used in the identity (2) (called *AZ identity*) in the following theorem.

Theorem 2 (Ahlswede and Zhang [3]). *If $\emptyset \neq \mathcal{A} \subseteq 2^{[n]}$ and $\emptyset \notin \mathcal{A}$ then*

$$\sum_{X \in \mathcal{U}(\mathcal{A})} \frac{|Z_{\mathcal{A}}(X)|}{|X|^{\binom{n}{|X|}}} = 1. \quad (2)$$

When $\mathcal{A}$ is an antichain, the identity (2) can be written as

$$\sum_{A \in \mathcal{A}} \frac{1}{\binom{n}{|A|}} + \sum_{X \in \mathcal{U}(\mathcal{A}) \setminus \mathcal{A}} \frac{|Z_{\mathcal{A}}(X)|}{|X|^{\binom{n}{|X|}}} = 1,$$

which is a sharpening of the LYM identity.

Theorem 3 (Bollobás [4]). *Let $q \in \mathbb{N}$, $\mathcal{A} = \{A_1, A_2, \dots, A_q\} \subseteq 2^{[n]}$ and $\mathcal{B} = \{B_1, B_2, \dots, B_q\} \subseteq 2^{[n]}$ such that $A_i \subseteq B_j$ if and only if $i = j$. Then*

$$\sum_{i=1}^q \frac{1}{\binom{n-|B_i|+|A_i|}{|A_i|}} \leq 1. \quad (3)$$

Ahlswede and Cai also use the AZ function to establish the identity (4) in the following theorem to sharpen the identity (3) of Bollobás for two set systems.

Theorem 4 (Ahlswede and Cai [1]). *Let $q \in \mathbb{N}$, $\mathcal{A} = \{A_1, A_2, \dots, A_q\} \subseteq 2^{[n]}$ and $\mathcal{B} = \{B_1, B_2, \dots, B_q\} \subseteq 2^{[n]}$ such that $A_i \subseteq B_j$ if and only if $i = j$. Then*

$$\sum_{i=1}^q \frac{1}{\binom{n-|B_i|+|A_i|}{|A_i|}} + \sum_{X \in \mathcal{U}(\mathcal{A}) \setminus \mathcal{D}(\mathcal{B})} \frac{|Z_{\mathcal{A}}(X)|}{|X|^{\binom{n}{|X|}}} = 1. \quad (4)$$

In [1], a generalized identity is introduced for two-set systems $\mathcal{A}, \mathcal{B}$ satisfying the more general condition: $\mathcal{B} \subseteq \mathcal{U}(\mathcal{A})$, and for each $A \in \mathcal{A}$, there exists uniquely $B \in \mathcal{B}$ such that $A \subset B$. When this condition holds, put $\mathcal{B} = \{B_1, B_2, \dots, B_q\}$ and $\mathcal{A}_i = \{A \in \mathcal{A} : A \subset B_i\}$ for $1 \leq i \leq q$. Then the identity is

$$\sum_{i=1}^q \sum_{\emptyset \neq S \subseteq \mathcal{A}_i} \frac{(-1)^{|S|-1}}{\binom{n-|B_i|+|\bigcup_{A \in S} A|}{|\bigcup_{A \in S} A|}} + \sum_{X \in \mathcal{U}(\mathcal{A}) \setminus \mathcal{D}(\mathcal{B})} \frac{|Z_{\mathcal{A}}(X)|}{|X|^{\binom{n}{|X|}}} = 1. \quad (5)$$

Note that $A \subseteq B_i$ for all $A \in \mathcal{A}_i$ and $A \not\subseteq B_j$ for $j \neq i$ and $A \in \mathcal{A}_i$.

The dual function of AZ function (called *AZ dual function*), the $Z^*(\cdot)$ is used in dual identities of AZ-style identities [5, 7, 12] (called *AZ-style identities*).

There are many contributions on generalization of the AZ identity [3], or establishing AZ-style identities for several posets of a specific kind or abstract posets [2, 5–7, 10, 12–15]. A motivation for this paper is to find a generic type for AZ-style identities on Boolean lattice. The application of this generic type is that it can be used as a template to establish AZ-style identities. The generic type also explains why there are many AZ-style identities to be found in recent years.

2. Main Theorems

Theorem 5. *If $f(x)$ is a real function defined on non-negative integers and $\emptyset \neq \mathcal{A} \subseteq 2^{[n]}$, then*

$$\sum_{X \in \mathcal{U}(\mathcal{A})} f(|X|)|Z_{\mathcal{A}}(X)| = \sum_{X \in \mathcal{U}(\mathcal{A})} [|X|f(|X|) - (n - |X|)f(|X| + 1)]. \quad (6)$$

Theorem 6. *If $f(x)$ is a real function defined on non-negative integers and $\mathcal{A}, \mathcal{B} \subseteq 2^{[n]}$, $\mathcal{A} \neq \emptyset \neq \mathcal{B}$, then*

$$\begin{aligned} & \sum_{X \in \mathcal{U}(\mathcal{A}) \setminus \mathcal{D}(\mathcal{B})} f(|X|)|Z_{\mathcal{A}}(X)| + \sum_{X \in \mathcal{U}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B})} f(|X| + 1)(n - |Z_{\mathcal{B}}^*(X)|) \\ &= \sum_{X \in \mathcal{U}(\mathcal{A}) \setminus \mathcal{D}(\mathcal{B})} [|X|f(|X|) - (n - |X|)f(|X| + 1)]. \end{aligned} \quad (7)$$

If $\mathcal{U}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B}) = \emptyset$ then (7) becomes (6). If $\emptyset \in \mathcal{A}$ then $\mathcal{U}(\mathcal{A}) = 2^{[n]}$, $Z_{\mathcal{A}}(X) = \emptyset$, $|Z_{\mathcal{A}}(X)| = 0$ for all $X \in \mathcal{U}(\mathcal{A})$, then (6) becomes an elementary combinatorial identity,

$$\sum_{k=0}^n k \binom{n}{k} f(k) = \sum_{k=0}^n (n - k) \binom{n}{k} f(k + 1),$$

which does not depend on the $Z_{\mathcal{A}}(\cdot)$ function; (7) becomes the dual of (6)

$$\sum_{X \in \mathcal{D}(\mathcal{B})} g(|X|)(n - |Z_{\mathcal{B}}^*(X)|) = \sum_{X \in \mathcal{D}(\mathcal{B})} [(n - |X|)g(|X|) - |X|g(|X| - 1)],$$

which is equivalent to

$$\sum_{X \in \mathcal{D}(\mathcal{B})} g(|X|)|Z_{\mathcal{B}}^*(X)| = \sum_{X \in \mathcal{D}(\mathcal{B})} [|X|g(|X|) + |X|g(|X| - 1)].$$

In fact, the two identities (6) and (7) are interesting when $\mathcal{U}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B}) \neq \emptyset$ and $\emptyset \notin \mathcal{A}$. We are going to prove these identities in the next subsection.

2.1. Proof of Theorems

Lemma 1 (See Proof of [13, Theorem 3.1]). *Let k be an integer such that $1 \leq k \leq n$, $\emptyset \neq \mathcal{A} \subseteq 2^{[n]}$. Then*

$$\sum_{X \in \mathcal{U}(\mathcal{A}) \cap \mathcal{P}_k} |Z_{\mathcal{A}}(X)| = k|\mathcal{U}(\mathcal{A}) \cap \mathcal{P}_k| - (n - k + 1)|\mathcal{U}(\mathcal{A}) \cap \mathcal{P}_{k-1}|. \quad (8)$$

Proof of Theorem 5. We write the left-hand side of (6) as

$$0 + \sum_{k=1}^n \sum_{X \in \mathcal{U}(\mathcal{A}) \cap \mathcal{P}_k} f(|X|)|Z_{\mathcal{A}}(X)| = \sum_{k=1}^n f(k) \sum_{X \in \mathcal{U}(\mathcal{A}) \cap \mathcal{P}_k} |Z_{\mathcal{A}}(X)|.$$

Then (8) is applied to obtain

$$\begin{aligned}
 \sum_{X \in \mathcal{U}(\mathcal{A})} f(|X|)|Z_{\mathcal{A}}(X)| &= \sum_{k=1}^n f(k) \sum_{X \in \mathcal{U}(\mathcal{A}) \cap \mathcal{P}_k} |Z_{\mathcal{A}}(X)| \\
 &= \sum_{k=1}^n f(k)k|\mathcal{U}(\mathcal{A}) \cap \mathcal{P}_k| - \sum_{k=1}^n f(k)(n-k+1)|\mathcal{U}(\mathcal{A}) \cap \mathcal{P}_{k-1}| \\
 &= \sum_{k=1}^n f(k)k|\mathcal{U}(\mathcal{A}) \cap \mathcal{P}_k| - \sum_{k=0}^{n-1} f(k+1)(n-k)|\mathcal{U}(\mathcal{A}) \cap \mathcal{P}_k| \\
 &= \sum_{X \in \mathcal{U}(\mathcal{A})} |X|f(|X|) - \sum_{X \in \mathcal{U}(\mathcal{A})} (n-|X|)f(|X|+1)
 \end{aligned}$$

which is the right-hand side of (6). $\square$

For $\emptyset \neq X \subseteq 2^{[n]}$, ΔX denotes the set of subsets $X \setminus \{\alpha\}$ with $\alpha \in X$; ∇X denotes the set of subsets $X \cup \{\beta\}$ of $[n]$ with $\beta \notin X$. Obviously, we have $|\Delta X| = |X|$ and $|\nabla X| = n - |X|$.

Lemma 2 (See [14, Lemma 3.1]). *Let $\emptyset \neq \mathcal{A} \subseteq 2^{[n]}$, $X \in \mathcal{U}(\mathcal{A})$, $Y \in \mathcal{D}(\mathcal{A})$. Then*

(i)

$$|X| = |Z_{\mathcal{A}}(X)| + |\Delta X \cap \mathcal{U}(\mathcal{A})|,$$

(ii)

$$|Y| = |Z_{\mathcal{A}}^*(Y)| - |\nabla Y \cap \mathcal{D}(\mathcal{A})|.$$

Lemma 3. *Let $\emptyset \neq \mathcal{A}, \mathcal{B} \subseteq 2^{[n]}$, k be an integer, $1 \leq k \leq n$. Then*

$$\begin{aligned}
 &\sum_{X \in \mathcal{U}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B}) \cap \mathcal{P}_k} |Z_{\mathcal{A}}(X)| + \sum_{X \in \mathcal{U}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B}) \cap \mathcal{P}_{k-1}} |Z_{\mathcal{B}}^*(X)| \\
 &= k|\mathcal{U}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B}) \cap \mathcal{P}_k| + (k-1)|\mathcal{U}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B}) \cap \mathcal{P}_{k-1}|.
 \end{aligned} \tag{9}$$

Proof. Put $\mathcal{F} = \mathcal{U}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B})$. The case $\mathcal{F} = \emptyset$ is trivial. Let us consider the case $\mathcal{F} \neq \emptyset$. The identity (9) is equivalent to following identity

$$\sum_{X \in \mathcal{F} \cap \mathcal{P}_k} [|X| - |Z_{\mathcal{A}}(X)|] = \sum_{Y \in \mathcal{F} \cap \mathcal{P}_{k-1}} [|Z_{\mathcal{B}}^*(Y)| - |Y|]. \tag{10}$$

Let δ_k be the number of pairs (X, Y) satisfying $X \supset Y$; $X, Y \in \mathcal{F}$, and $|X| = k, |Y| = k-1$. To prove (10), we count δ_k in two ways.

For all $X \in \mathcal{F}$ and $|X| = k$, by using Lemma 2, $|X| - |Z_{\mathcal{A}}(X)|$ is the number of $\alpha \in X$ such that $X \setminus \{\alpha\} \in \mathcal{U}(\mathcal{A})$. Since $X \setminus \{\alpha\} \in \mathcal{D}(\mathcal{B})$, $|X| - |Z_{\mathcal{A}}(X)|$ is also the

number of $\alpha \in X$ such that $X \setminus \{\alpha\} \in \mathcal{F}$. Hence the left-hand side of (10) equals to δ_k .

Similarly, for all $Y \in \mathcal{F}$ and $|Y| = k - 1$, $|Z_{\mathcal{B}}^*(Y)| - |Y|$ is the number of $\beta \notin Y$ such that $Y \cup \{\beta\} \in \mathcal{F}$. Therefore, the right-hand side of (10) equals to δ_k as well. So both two sides of (10) equal to δ_k . $\square$

Proof of Theorem 6. Multiply by $f(k)$ on both sides of (9) to get

$$\begin{aligned} & \sum_{X \in \mathcal{U}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B}) \cap \mathcal{P}_k} f(|X|) |Z_{\mathcal{A}}(X)| + \sum_{X \in \mathcal{U}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B}) \cap \mathcal{P}_{k-1}} f(|X| + 1) |Z_{\mathcal{B}}^*(X)| \\ &= \sum_{X \in \mathcal{U}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B}) \cap \mathcal{P}_k} |X| f(|X|) + \sum_{X \in \mathcal{U}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B}) \cap \mathcal{P}_{k-1}} |X| f(|X| + 1). \end{aligned}$$

Take $k = 1, 2, \dots, n$ in the identity (11), then add these identities and consider some trivial cases of $\mathcal{U}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B}) \cap \mathcal{P}_0$ and $\mathcal{U}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B}) \cap \mathcal{P}_n$, we obtain the following corollary.

Corollary 1.

$$\begin{aligned} & \sum_{X \in \mathcal{U}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B})} f(|X|) |Z_{\mathcal{A}}(X)| + \sum_{X \in \mathcal{U}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B})} f(|X| + 1) |Z_{\mathcal{B}}^*(X)| \\ &= \sum_{X \in \mathcal{U}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B})} |X| f(|X|) + \sum_{X \in \mathcal{U}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B})} |X| f(|X| + 1). \end{aligned} \quad (11)$$

Subtracting $\sum_{X \in \mathcal{U}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B})} n f(|X| + 1)$ from each side of (11) we have

$$\begin{aligned} & \sum_{X \in \mathcal{U}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B})} f(|X|) |Z_{\mathcal{A}}(X)| - \sum_{X \in \mathcal{U}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B})} f(|X| + 1) (n - |Z_{\mathcal{B}}^*(X)|) \\ &= \sum_{X \in \mathcal{U}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B})} [|X| f(|X|) - (n - |X|) f(|X| + 1)]. \end{aligned} \quad (12)$$

Finally, from the identity (6) of Theorem 5 and (12), and noticing that $\mathcal{U}(\mathcal{A}) \setminus \mathcal{D}(\mathcal{B}) = \mathcal{U}(\mathcal{A}) \setminus [\mathcal{U}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B})]$, we deduce immediately the identity (7) of Theorem 6. $\square$

2.2. More generalized form of the AZ identity for two-set systems

The AZ identity for two-set systems, the identity (5), holds for two families $\emptyset \neq \mathcal{A}$, $\mathcal{B} \subseteq 2^{[n]}$ satisfying $\mathcal{A} = \bigcup_{i=1}^q \mathcal{A}_i$; $\mathcal{B} = \{B_1, B_2, \dots, B_q\}$; $A \subseteq B_i$ for all $A \in \mathcal{A}_i$; $A \not\subseteq B_j$ for $i \neq j$, $A \in \mathcal{A}_i$. The identity is used to study extremal cases of some important inequalities, which were considered very deep in [1, 11].

The first term on the left-hand side of (5) is fairly complex. We tried to analyze elements participating to the quantity to understand what they are. The following theorem shows more clear this quantity.

Theorem 7. If $\emptyset \neq \mathcal{A}, \mathcal{B} \subseteq 2^{[n]}$ and $[n] \notin \mathcal{B}$ then

$$\sum_{X \in \mathcal{U}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B})} \frac{n - |Z_{\mathcal{B}}^*(X)|}{(n - |X|) \binom{n}{|X|}} + \sum_{X \in \mathcal{U}(\mathcal{A}) \setminus \mathcal{D}(\mathcal{B})} \frac{|Z_{\mathcal{A}}(X)|}{|X| \binom{n}{|X|}} = 1. \quad (13)$$

Proof. Notice that $[n] \notin \mathcal{B}$ and $\emptyset \in \mathcal{D}(\mathcal{B})$ imply $[n] \notin \mathcal{U}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B})$, $\emptyset \notin \mathcal{U}(\mathcal{A}) \setminus \mathcal{D}(\mathcal{B})$. In the identity (7), take $f(x) = \frac{1}{x \binom{n}{x}}$ if $1 \leq x \leq n$ and $f(x) = 0$ otherwise, we have

$$\begin{aligned} & \sum_{X \in \mathcal{U}(\mathcal{A}) \setminus \mathcal{D}(\mathcal{B})} \frac{|Z_{\mathcal{A}}(X)|}{|X| \binom{n}{|X|}} + \sum_{X \in \mathcal{U}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B})} \frac{n - |Z_{\mathcal{B}}^*(X)|}{(|X| + 1) \binom{n}{|X|+1}} \\ &= 1 + \sum_{X \in \mathcal{U}(\mathcal{A}) \setminus \mathcal{D}(\mathcal{B}), |X| < n} \left[|X| \frac{1}{|X| \binom{n}{|X|}} - (n - |X|) \frac{1}{(|X| + 1) \binom{n}{|X|+1}} \right] \\ &= 1 + \sum_{X \in \mathcal{U}(\mathcal{A}) \setminus \mathcal{D}(\mathcal{B}), |X| < n} \left[\frac{1}{\binom{n}{|X|}} - \frac{n - |X|}{(|X| + 1) \binom{n}{|X|+1}} \right]. \end{aligned}$$

Since $(k + 1) \binom{n}{k+1} = (n - k) \binom{n}{k}$ for $0 \leq k < n$, if $|X| < n$ then

$$\frac{n - |Z_{\mathcal{B}}^*(X)|}{(|X| + 1) \binom{n}{|X|+1}} = \frac{n - |Z_{\mathcal{B}}^*(X)|}{(n - |X|) \binom{n}{|X|}} \quad \text{and} \quad \left[\frac{1}{\binom{n}{|X|}} - \frac{n - |X|}{(|X| + 1) \binom{n}{|X|+1}} \right] = 0.$$

Hence we deduce immediately the identity (13). $\square$

Notes. The identity (13) involves the other AZ identities and even their duals. If $\mathcal{U}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B}) = \emptyset$ which implies $\emptyset \notin \mathcal{A}$ (because of $\emptyset \in \mathcal{D}(\mathcal{B})$), then the identity (13) becomes the original AZ identity $\sum_{X \in \mathcal{U}(\mathcal{A})} \frac{|Z_{\mathcal{A}}(X)|}{|X| \binom{n}{|X|}} = 1$. If $\emptyset \in \mathcal{A}$ then $\mathcal{U}(\mathcal{A}) = 2^{[n]}$ and $Z_{\mathcal{A}}(X) = \emptyset$, and we get the dual of the AZ identity $\sum_{X \in \mathcal{D}(\mathcal{B})} \frac{n - |Z_{\mathcal{B}}^*(X)|}{(n - |X|) \binom{n}{|X|}} = 1$. In fact, the identity (13) is meaningful for two-set systems $\mathcal{A}, \mathcal{B}$ only if $\mathcal{U}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B}) \neq \emptyset$, $\emptyset \notin \mathcal{A}$, $[n] \notin \mathcal{B}$.

Notice that the condition $[n] \notin \mathcal{B}$ is weaker than that of Ahlswede–Cai for the identity (5). When the Ahlswede–Cai condition satisfies, we deduce the following from (5) and (13)

$$\sum_{X \in \mathcal{U}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B})} \frac{n - |Z_{\mathcal{B}}^*(X)|}{(n - |X|) \binom{n}{|X|}} = \sum_{i=1}^q \sum_{\emptyset \neq S \subseteq \mathcal{A}_i} \frac{(-1)^{|S|-1}}{\binom{n - |B_i| + |\bigcup_{A \in S} A|}{|\bigcup_{A \in S} A|}}.$$

3. Parametrized AZ Identities

The original AZ identity can be extended to a parametrized AZ-style identity $\sum_{X \in \mathcal{U}(\mathcal{A})} \frac{|Z_{\mathcal{A}}(X)| + m}{(|X| + m) \binom{n+m}{|X|+m}} = 1$ [12], where m is an integer such that $|A| + m > 0$ for each $A \in \mathcal{A}$ satisfying $\mathcal{A} \neq \emptyset$, $\emptyset \notin \mathcal{A} \subseteq 2^{[n]}$.

Moreover, when x is a real number and k is a non-negative integer, the binomial coefficient $\binom{x}{k}$ can be defined by $\frac{x(x-1)\cdots(x-k+1)}{k!}$, and if y is a real number so that $x - y$ is a non-negative integer, $\binom{x}{y} = \binom{x}{x-y}$. The AZ identity is true for all values of m that the denominators are not zero [15], $m \in \mathbb{R} \setminus \{-r, -(r+1), \dots, -n\}$, where $r = \min\{|A|/A \in \mathcal{A}\}$. This is a continuous version of AZ identities. For example by taking $m = -\pi$, we can get an identity such as $\sum_{X \in \mathcal{U}(\mathcal{A})} \frac{|Z_{\mathcal{A}}(X)| - \pi}{(|X| - \pi) \binom{n - \pi}{|X| - \pi}} = 1$, even though π is a transcendent number. In fact, we can also assume $m \in \mathbb{C} \setminus \{-r, -(r+1), \dots, -n\}$.

In this section, we investigate the case of AZ identities parametrized by real numbers.

Lemma 4. *Let $f(x)$ be a real function defined on non-negative integers, m be a real number and $\mathcal{A}, \mathcal{B} \subseteq 2^{[n]}$, $\mathcal{A} \neq \emptyset \neq \mathcal{B}$. Then*

(a)

$$\begin{aligned} & \sum_{X \in \mathcal{U}(\mathcal{A})} f(|X|)(m + |Z_{\mathcal{A}}(X)|) \\ &= \sum_{X \in \mathcal{U}(\mathcal{A})} [(m + |X|)f(|X|) - (n - |X|)f(|X| + 1)]. \end{aligned} \quad (14)$$

(b)

$$\begin{aligned} & \sum_{X \in \mathcal{U}(\mathcal{A}) \setminus \mathcal{D}(\mathcal{B})} f(|X|)(m + |Z_{\mathcal{A}}(X)|) + \sum_{X \in \mathcal{U}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B})} f(|X| + 1)(n - |Z_{\mathcal{B}}^*(X)|) \\ &= \sum_{X \in \mathcal{U}(\mathcal{A}) \setminus \mathcal{D}(\mathcal{B})} [(m + |X|)f(|X|) - (n - |X|)f(|X| + 1)]. \end{aligned} \quad (15)$$

(c)

$$\begin{aligned} & \sum_{X \in \mathcal{D}(\mathcal{B})} f(|X| + 1)(m - |Z_{\mathcal{B}}^*(X)|) \\ &= \sum_{X \in \mathcal{D}(\mathcal{B})} [(m - |X|)f(|X| + 1) - |X|f(|X|)]. \end{aligned} \quad (16)$$

(d)

$$\begin{aligned} & \sum_{X \in \mathcal{D}(\mathcal{B}) \setminus \mathcal{U}(\mathcal{A})} f(|X| + 1)(m - |Z_{\mathcal{B}}^*(X)|) + \sum_{X \in \mathcal{U}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B})} f(|X|)|Z_{\mathcal{A}}(X)| \\ &= \sum_{X \in \mathcal{D}(\mathcal{B}) \setminus \mathcal{U}(\mathcal{A})} [(m - |X|)f(|X| + 1) - |X|f(|X|)]. \end{aligned} \quad (17)$$

Proof. Using Corollary 1, subtracting a suitable quantity from each side of the identity (11) we get immediately the identities of Lemma 4. $\square$

Notes. Almost all properties of the binomial coefficient are still correct for the case of real numbers. The property $(k+1)\binom{x}{k+1} = (x-k)\binom{x}{k}$ is true for $x \in \mathbb{R}$, and $k \in \mathbb{N}$ or $x-k \in \mathbb{N}$, which is used in the proof of some theorems. In fact, the property is interpreted as

$$\frac{1}{\binom{x}{k}} - \frac{x-k}{(k+1)\binom{x}{k+1}} = 0 \quad \text{if } x \geq k+1.$$

Theorem 8. Let $\mathcal{A}, \mathcal{B} \subseteq 2^{[n]}$, $\mathcal{A} \neq \emptyset \neq \mathcal{B}$, $r = \min\{|A|/A \in \mathcal{A}\}$, $q = \max\{|B|/B \in \mathcal{B}\}$, m be a real number.

(a) If $[n] \notin \mathcal{B}$ and $m \notin \{-r, -(r+1), \dots, -n\}$ then

$$\sum_{X \in \mathcal{U}(\mathcal{A}) \setminus \mathcal{D}(\mathcal{B})} \frac{m + |Z_{\mathcal{A}}(X)|}{(m + |X|)\binom{m+n}{m+|X|}} + \sum_{X \in \mathcal{U}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B})} \frac{n - |Z_{\mathcal{B}}^*(X)|}{(n - |X|)\binom{m+n}{m+|X|}} = 1. \quad (18)$$

(b) If $\emptyset \notin \mathcal{A}$ and $m \notin \{0, 1, \dots, q\}$ then

$$\sum_{X \in \mathcal{D}(\mathcal{B}) \setminus \mathcal{U}(\mathcal{A})} \frac{m - |Z_{\mathcal{B}}^*(X)|}{(m - |X|)\binom{m}{|X|}} + \sum_{X \in \mathcal{U}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B})} \frac{|Z_{\mathcal{A}}(X)|}{|X|\binom{m}{|X|}} = 1. \quad (19)$$

Notice that we always have $[n] \in \mathcal{U}(\mathcal{A})$ and $\emptyset \in \mathcal{D}(\mathcal{B})$. Therefore, if $[n] \notin \mathcal{B}$ then $[n] \notin \mathcal{U}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B})$ which implies $n - |X| \neq 0$ for $X \in \mathcal{U}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B})$; if $\emptyset \notin \mathcal{A}$ then $\emptyset \notin \mathcal{U}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B})$, and $|X| \neq 0$ for $X \in \mathcal{U}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B})$.

For all $m \in \mathbb{R}$, the terms $\binom{m+n}{m+|X|}$ and $\binom{m}{|X|}$ are always defined because of $(m+n) - (m+|X|) = n - |X| \in \mathbb{N}$ and $|X| \in \mathbb{N}$. The condition $m \notin \{-r, -(r+1), \dots, -n\}$ insures that the terms $(m+|X|)\binom{m+n}{m+|X|}$ are different from zero. Similarly, the terms $(m-|X|)\binom{m}{|X|}$ are not zero when $m \notin \{0, 1, \dots, q\}$.

Proof. To prove (a), we use the identity (15) of Lemma 4. Take $f(x) = \frac{1}{(m+x)\binom{m+n}{m+x}}$ if $x \in \{r, r+1, \dots, n\}$ and $f(x) = 0$ otherwise, we have

$$\begin{aligned} & \sum_{X \in \mathcal{U}(\mathcal{A}) \setminus \mathcal{D}(\mathcal{B})} \frac{m + |Z_{\mathcal{A}}(X)|}{(m + |X|)\binom{m+n}{m+|X|}} + \sum_{X \in \mathcal{U}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B})} \frac{n - |Z_{\mathcal{B}}^*(X)|}{(m + |X| + 1)\binom{m+n}{m+|X|+1}} \\ &= \sum_{X \in \mathcal{U}(\mathcal{A}) \setminus \mathcal{D}(\mathcal{B}), |X| < n} [(m + |X|)f(|X|) - (n - |X|)f(|X| + 1)] \\ & \quad + [(m + n)f(n) - (n - n)f(n + 1)] \\ &= 1 + \sum_{X \in \mathcal{U}(\mathcal{A}) \setminus \mathcal{D}(\mathcal{B}), |X| < n} [(m + |X|)f(|X|) - (n - |X|)f(|X| + 1)]. \end{aligned}$$

Since $\frac{1}{(m+|X|+1)\binom{m+n}{m+|X|+1}} = \frac{1}{(n-|X|)\binom{m+n}{m+|X|}}$ for $0 \leq |X| < n$, we have $(m + |X|)f(|X|) - (n - |X|)f(|X| + 1) = 0$ and $\frac{n - |Z_{\mathcal{B}}^*(X)|}{(m+|X|+1)\binom{m+n}{m+|X|+1}} = \frac{n - |Z_{\mathcal{B}}^*(X)|}{(n-|X|)\binom{m+n}{m+|X|}}$ whenever

$0 \leq |X| < n$. Notice that $[n] \notin \mathcal{U}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B})$, and the sum over $\mathcal{U}(\mathcal{A}) \cap \mathcal{D}(\mathcal{B})$ does not contain the term $X = [n]$. Therefore, we can deduce the identity (18).

Similarly, to prove (b) we choose $f(x) = \frac{1}{x \binom{m}{x}}$ for $x \in \{1, 2, \dots, q+1\}$ and $f(x) = 0$ otherwise. Then $f(x+1) = \frac{1}{(x+1) \binom{m}{x+1}} = \frac{1}{(m-x) \binom{m}{x}}$ for $x \in \{0, 1, \dots, q\}$ and $f(x+1) = 0$ otherwise. We derive the identity (19) from the identity (17) of Lemma 4. $\square$

4. Other Instances of Generic Identity

4.1. Elementary and simple cases

When $f(x) = 0$ for all $x \neq n$ and $f(n) = 1$, the identity (6) gives

$$|Z_{\mathcal{A}}([n])| = (0 - |\mathcal{U}(\mathcal{A}) \cap \mathcal{P}_{n-1}|) + (n - 0)$$

which is $|\mathcal{U}(\mathcal{A}) \cap \mathcal{P}_{n-1}| + |\bigcap_{A \in \mathcal{A}} A| = n$.

For next corollary, take $f(x) = 1$ for all x , we have

$$\sum_{X \in \mathcal{U}(\mathcal{A})} |Z_{\mathcal{A}}(X)| = \sum_{X \in \mathcal{U}(\mathcal{A})} |X| - \sum_{X \in \mathcal{U}(\mathcal{A})} (n - |X|)$$

which is equivalent to

$$\sum_{X \in \mathcal{U}(\mathcal{A})} |Z_{\mathcal{A}}(X)| = 2 \sum_{X \in \mathcal{U}(\mathcal{A})} |X| - n|\mathcal{U}(\mathcal{A})|. \quad (20)$$

Let $\mathcal{A}_0$ be the set of minimal sets of $\mathcal{A}$. Then $\mathcal{A}_0$ is an antichain, and $\mathcal{U}(\mathcal{A}) = \mathcal{U}(\mathcal{A}_0)$, $Z_{\mathcal{A}_0}(X) = Z_{\mathcal{A}}(X)$ for all $X \in \mathcal{U}(\mathcal{A})$, $Z_{\mathcal{A}}(A) = A$ for all $A \in \mathcal{A}_0$. Put $\mathcal{U} = \mathcal{U}(\mathcal{A}) = \mathcal{U}(\mathcal{A}_0)$. The identity (20) can be rewritten as

$$\frac{1}{|\mathcal{U}|} \sum_{X \in \mathcal{U}} |X| = \frac{n}{2} + \frac{\sum_{A \in \mathcal{A}_0} |A|}{2|\mathcal{U}|} + \frac{\sum_{X \in \mathcal{U} \setminus \mathcal{A}_0} |Z_{\mathcal{A}}(X)|}{2|\mathcal{U}|}. \quad (21)$$

The upset $\mathcal{U}(\mathcal{A})$ can be considered as a simple graph with edges (X, Y) , where $Y \supset X, |Y \Delta X| = 1$. For each vertex $X \in \mathcal{U}(\mathcal{A})$, the quantity $|Z_{\mathcal{A}}(X)|$ is the number of edges which leave $\mathcal{U}(\mathcal{A})$ (see [3]) from the vertex X . The identity (20) shows a relationship between the number of leaving edges and the volume of $\mathcal{U}(\mathcal{A})$. The identity (21) implies a classical result, *the average size of the sets in an upset is at least $\frac{n}{2}$* , i.e., $\frac{1}{|\mathcal{U}|} \sum_{X \in \mathcal{U}} |X| \geq \frac{n}{2}$. The equality happens whenever $\emptyset \in \mathcal{A}$ which is equivalent to $\mathcal{U}(\mathcal{A}) = 2^{[n]}$.

4.2. Original AZ identity and local AZ identities

Let $\emptyset \neq \mathcal{A} \subseteq 2^{[n]}$, $\emptyset \notin \mathcal{A}$, k be an integer, $1 \leq k \leq n$. When $f(x) = \frac{1}{x \binom{n}{x}}$ if $1 \leq x \leq n$ and $f(x) = 0$ otherwise, we derive from the identity (6) that

$$\sum_{X \in \mathcal{U}(\mathcal{A})} \frac{|Z_{\mathcal{A}}(X)|}{|X| \binom{n}{|X|}} = 1 + \sum_{X \in \mathcal{U}(\mathcal{A}) \setminus \{[n]\}} \left[\frac{1}{\binom{n}{|X|}} - \frac{n - |X|}{(|X| + 1) \binom{n}{|X| + 1}} \right] = 1.$$

Similarly, for the case of the poset $\mathcal{P}_{\leq k} = \bigcup_{1 \leq i \leq k} \mathcal{P}_i$ we can choose $f(x) = \frac{1}{x \binom{n}{x}}$ if $1 \leq x \leq k$ and $f(x) = 0$ otherwise, to get the identity of partial sums

$$\sum_{X \in \mathcal{U}(\mathcal{A}), |X| \leq k} \frac{|Z_{\mathcal{A}}(X)|}{|X| \binom{n}{|X|}} = \frac{|\mathcal{U}(\mathcal{A}) \cap \mathcal{P}_k|}{\binom{n}{k}}.$$

From the identity (6), if $f(x) = 0$ for all $x \neq k$, $f(k) = 1$ then

$$\sum_{X \in \mathcal{U}(\mathcal{A}) \cap \mathcal{P}_k} |Z_{\mathcal{A}}(X)| = k|\mathcal{U}(\mathcal{A}) \cap \mathcal{P}_k| - (n - k + 1)|\mathcal{U}(\mathcal{A}) \cap \mathcal{P}_{k-1}|,$$

which is the local AZ identity (see [13])

$$\frac{|\mathcal{U}(\mathcal{A}) \cap \mathcal{P}_{k-1}|}{\binom{n}{k-1}} + \sum_{X \in \mathcal{U}(\mathcal{A}) \cap \mathcal{P}_k} \frac{|Z_{\mathcal{A}}(X)|}{|X| \binom{n}{|X|}} = \frac{|\mathcal{U}(\mathcal{A}) \cap \mathcal{P}_k|}{\binom{n}{k}}.$$

4.3. Identities involving cardinalities of an upset or a downset

From the identity (14) of Lemma 4, we take $m = -(n+1)$ and $f(x) = \frac{1}{n+1-x}$ if $0 \leq x \leq n$; $f(x) = 0$ otherwise. Then its right-hand side is

$$\begin{aligned} & \left[\frac{-n-1+n}{n+1-n} + 0 \right] + \sum_{X \in \mathcal{U}(\mathcal{A}) \setminus \{[n]\}} \left[\frac{-n-1+|X|}{n+1-|X|} - \frac{n-|X|}{n+1-|X|-1} \right] \\ &= -1 + \sum_{X \in \mathcal{U}(\mathcal{A}) \setminus \{[n]\}} (-2) = -1 - 2[|\mathcal{U}(\mathcal{A})| - 1] = -2|\mathcal{U}(\mathcal{A})| + 1. \end{aligned}$$

Therefore, we obtain

$$\sum_{X \in \mathcal{U}(\mathcal{A})} \frac{n+1-|Z_{\mathcal{A}}(X)|}{n+1-|X|} = 2|\mathcal{U}(\mathcal{A})| - 1.$$

Similarly for the dual case, we can derive easily the identity

$$\sum_{X \in \mathcal{D}(\mathcal{A})} \frac{1+|Z_{\mathcal{A}}^*(X)|}{1+|X|} = 2|\mathcal{D}(\mathcal{A})| - 1$$

from the identity (16).

The last two identities were also investigated in [14].

Acknowledgment

The paper is supported by Vietnam National University, Ho Chi Minh City; Project No. HS2014-18-03.

References

- [1] R. Ahlswede and N. Cai, A generalisation of the AZ identity, *Combinatorica* **13** (1993) 241–247.
- [2] R. Ahlswede and N. Cai, Incomparability and intersection properties of Boolean interval lattices and chain posets, *European J. Combin.* **17** (1996) 677–687.

- [3] R. Ahlswede and Z. Zhang, An identity in combinatorial extremal theory, *Adv. Math.* **80** (1990) 137–151.
- [4] B. Bollobás, On generalized graphs, *Acta Math. Acad. Sci. Hungar.* **16** (1965) 447–452.
- [5] D. E. Daykin and T. D. Thu, The dual of Ahlswede–Zhang identity, *J. Combin. Theory Ser. A* **68** (1994) 246–249.
- [6] P. L. Erdős and L. A. Székely, Pseudo-LYM inequalities and AZ identities, *Adv. Appl. Math.* **19** (1997) 431–443.
- [7] A. J. W. Hilton and D. S. G. Stirling, The dual of an identity of Ahlswede and Zhang, *Congr. Numer.* **130** (1998) 113–126.
- [8] D. Lubell, A short proof of Sperners theorem, *J. Combin. Theory* **1** (1966) 209–214.
- [9] L. D. Meshalkin, Generalization of Sperners theorem on the number of subsets of a finite set, *Theory Probab. Appl.* **8** (1963) 203–204.
- [10] B. Patkós, AZ-type identities, http://web.ceu.hu/math/Research/Preprints_Archives/Patkos3.pdf.
- [11] A. J. Radcliffe and Z. Szaniszló, Extremal cases of the Ahlswede–Cai inequality, *J. Combin. Theory Ser. A* **76** (1996) 108–120.
- [12] T. D. Thu, An AZ-style identity and Bollobas deficiency, *J. Combin. Theory Ser. A* **114**(8) (2007) 1504–1514.
- [13] T. D. Thu, On local LYM identities, *Ann. Combin.* **17**(4) (2013) 755–763.
- [14] T. D. Thu, AZ-style identities of downsets and upsets, *Discrete Math. Algorithms Appl.* **6**(3) (2014) 1450040-1–8.
- [15] T. Thai-Duong and T. D. Thu, A simple proof for a more generalized form of the AZ-identity, *Vietnam J. Math.* **43** (2015) 771–775.
- [16] K. Yamamoto, Logarithmic order of free distributive lattice, *J. Math. Soc. Japan* **6** (1954) 343–353.